

Yeaseen Arafat

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RESEARCH INTERESTS

My research focuses on automated validation of program translation to ensure security of transformed software.

EDUCATION

🎓 The University of Utah

Ph.D. in Computer Science

Aug 2023 – Dec 2028 (Expected)

M.S. in Computer Science | CGPA: 4.00/4.00

Aug 2023 – Dec 2025

🎓 Bangladesh University of Engineering and Technology (BUET)

B.Sc. Engg. in Computer Science and Engineering | CGPA: 3.49/4.00

Feb 2015 – Apr 2019

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

May 2024 – Present

FuTURES³ Lab, The University of Utah.

Utah, USA

- Conducting research on automated cross-language program validation.
- Developed **PROGnosticator**, a construct-oriented transpiler fuzzer using LLM-assisted program reasoning and generation, discovering 64 new bugs—62 confirmed by developers—across seven C, Go, and JS transpilers.
- Designed and implemented **TeTRIS**, a general-purpose mutational fuzzing framework that discovered 12 previously unknown transpiler bugs, all confirmed by developers, across seven C, Go, and Haxe transpilers.
- Developed a cross-language parser that normalizes C, Go, and Haxe into a neutral Abstract Syntax Tree with scope and type inference, enabling syntax- and semantics-preserving program mutations.

Lecturer

June 2019 – July 2023(On Leave)

Department of Computer Science and Engineering, Southeast University

Dhaka, Bangladesh

- Taught core computer science courses and supervised undergraduate theses in machine learning and data analysis, contributing to curriculum design and program modernization.

Undergraduate Student Researcher

Apr 2018 – Apr 2019

Bangladesh University of Engineering and Technology

- Designed a machine learning-based intrusion detection system for the WiFi MAC layer (IEEE 802.11), implemented in Network Simulator-2, using statistical network traffic features to identify malicious behavior.

PUBLICATIONS

- **Yeaseen Arafat**, Stefan Nagy, “PROGnosticator: Testing Source-to-Source Code Translators via Construct-oriented Fuzzing”, *ACM International Conference on the Foundations of Software Engineering (FSE 2026)*. [pdf]
- **Yeaseen Arafat**, Stefan Nagy, “TeTRIS: General-purpose Fuzzing for Translation Bugs in Source-to-Source Code Transpilers”, *Annual Computer Security Applications Conference (ACSAC 2025)*. [doi] [pdf] [code]
- **Yeaseen Arafat**, Kazi Samin Yeaser, Ashikur Rahman, Arnab Dasgupta, “A Machine Learning based Approach for Protecting Wireless Networks Against DoS Attacks”, *International Conference on Networking, Systems and Security (NSysS 2020)*. [doi] [pdf] [code]

OPEN SOURCE CONTRIBUTIONS

- **Bug Discovery:** Discovered **80+** developer-confirmed previously-unknown bugs across projects such as c2rust, Go compiler, Zig, cxgo, swc and go2hx. [FuTURES³ Lab’s Bug Tracker]
- **Upstream Contributions:** Merged PRs to Node.js, llvm-project, and C2Rust on filesystem correctness, test determinism, and language semantics, with additional fixes in Polyglot and cxgo. [details]
- **TeTRIS:** Publicly reviewed and discussed by the Go2Hx developer community. [details]
- **Security Disclosure:** Identified a critical key-exposure vulnerability in a mobile banking app. [details]

TALKS AND PRESENTATIONS

- **Paper:** *TeTRIS: General-purpose Fuzzing for Translation Bugs in Source-to-Source Code Transpilers*. Annual Computer Security Applications Conference (ACSAC 2025), Hawaii, Dec 8–12, 2025. [\[slides\]](#)
- **Poster:** *Fuzzing for Code Translation Bugs in Source-to-Source Code Transpilers*. Kahlert School of Computing Graduate Visit Day, University of Utah, Feb 21–22, 2025. [\[talk\]](#) [\[poster\]](#)
- **Paper:** *A Machine Learning based Approach for Protecting Wireless Networks Against DoS Attacks*. International Conference on Networking, Systems and Security (NSysS 2020). [\[talk\]](#) [\[slides\]](#)

TECHNICAL PROJECTS

CyberShield-Spectrum | *Virtualized Security Lab*

 [GitHub](#)

- Built and tested a virtual enterprise network with VPN, firewall, IDS, and jump server configurations.
- Performed penetration testing and threat hunting using Snort, Volatility3, and FTK Imager.

KerFuzz: Linux Kernel Fuzzer | *C, Bash, GDB, Syzkaller, QEMU*

 [GitHub](#)

- Designed a syscall-level kernel fuzzing workflow using Syzkaller to discover bugs in the Linux kernel.
- Instrumented and analyzed faulting executions using GDB and custom Bash scripts for postmortem triage.

EduCComp | *Flex, Bison, YACC*

 [GitHub](#)

- Built a compiler for a C-subset, including Symbol Table, Lexical, Syntax, and Semantic Analysis.
- Implemented an Intermediate Code Generator, translating source code into x86 assembly.

Machine Learning projects | *Python*

 [GitHub](#)

- Implemented ML algorithms from scratch, including AdaBoost with Decision Trees, Perceptron, and ANN.
- Built recommender (Matrix Factorization), unsupervised models (PCA, GMM), and Channel Equalizer.

IInstagram | *MERN Stack*

 [GitHub](#)

- Built an Instagram-like web app with backend APIs (Node.js/Express), JWT auth, and image uploads (Cloudinary).
- Implemented password recovery with Twilio email integration, along with React Hooks, Router, and Context API for state management and group chat using Socket.IO.

AWARDS & SCHOLARSHIP

- **Graduate Research Fellowship**, University of Utah (Aug–Dec 2023)
- **University Merit Scholarship**, BUET (2015–2019)
- **Junior Government Scholarship**, Chittagong Education Board, Bangladesh (2009)

OUTREACH ACTIVITIES

- **Artifact Evaluation Committee Member:** [ACM CCS 2024](#), ACSAC 2025
- **Graduate Teaching Assistant:** CSE-4480 (Computer Networks), University of Utah (Spring 2024). Supported 170+ students via office hours, grading, and final evaluations.
- **Organizing Committee Member:** ACM ICPC ASIA Regional, Dhaka Site 2019
- **Member:** ACM; Cybersecurity Club, University of Utah [\[utahsec\]](#)

TECHNICAL SKILLS

Programming & Systems: C/C++, Rust, LLVM IR, Go, Python, Java, Assembly (x86, MIPS), Wasm

Security & Fuzzing: AFL++, Syzkaller, QEMU, Burp Suite, Wireshark, sqlmap, Snort, Ghidra, Pwndbg

Compilers & Analysis: LLVM, Flex/Bison/YACC, Nachos, Make, Static Analysis

Machine Learning: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy

Web & DevOps: React, Node.js, Express, Postman, Docker, GitHub Actions, GraphQL, REST API

REFERENCES

Dr. Stefan Nagy

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Kahlert School of Computing
The University of Utah
PhD Supervisor
Email: snagy@cs.utah.edu

Dr. A.K.M. Ashikur Rahman

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